HOLY CROSS COLLEGE

Life to the Full

11 Physics

Electricity

ELECTROSTATICS

If two objects of different materials are rubbed together, such as a plastic ruler and a woollen jumper, static electricity results. The friction between the objects causes a transfer of electrons from one of the materials to the other. This results in the bodies acquiring equal and opposite static charges.

Electrostatics is the study of the causes and effects of these static charges. Some common occurrences that can be explained by electrostatics are listed.

- You can sometimes get a small electrical shock after walking across new carpet.
- A plastic comb run through your hair will attract small pieces of paper.
- A balloon that is rubbed against some clothing will tend to stick to walls.
- You can sometimes feel a small electrical shock from a car after travelling for some time.

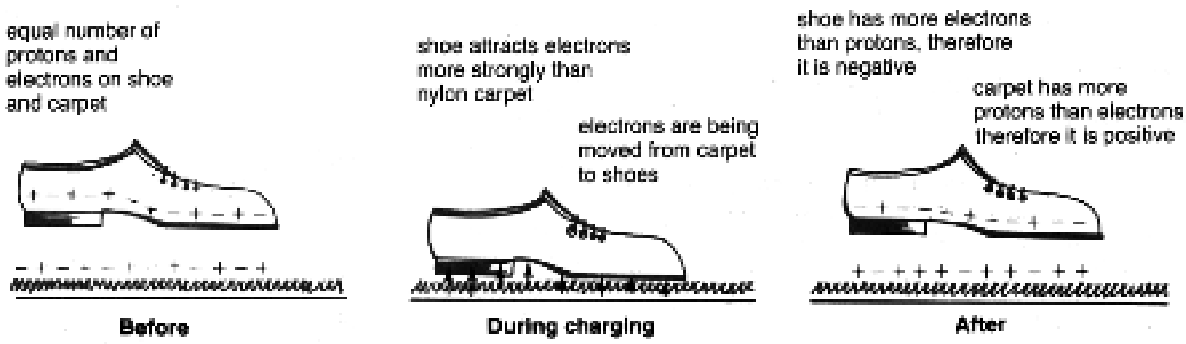
Electrostatic experiments with many different materials have established the following.

- Only two types of charges exist, positive and negative. ***Positively charged objects have a deficiency of electrons. Negatively charged objects have an excess of electrons.***
- ***Like charges repel, unlike charges attract*** (called the ***Charge Law***).
- Both positively and negatively charged objects attract neutral conductors.

The build up of charge on an object is called ***static electricity***.

Examples

1. Consider the build up of charge when walking on certain carpets.



2. ***Formation of Lightning***

If the charge build up on two objects is great enough, and these charges are opposite in sign, then a spark may jump between the two.

It is this effect that is responsible for lightning discharging between clouds and the Earth.

Strong updrafts within the clouds drive ice crystals and hailstones into collisions, forming a separation of charge.

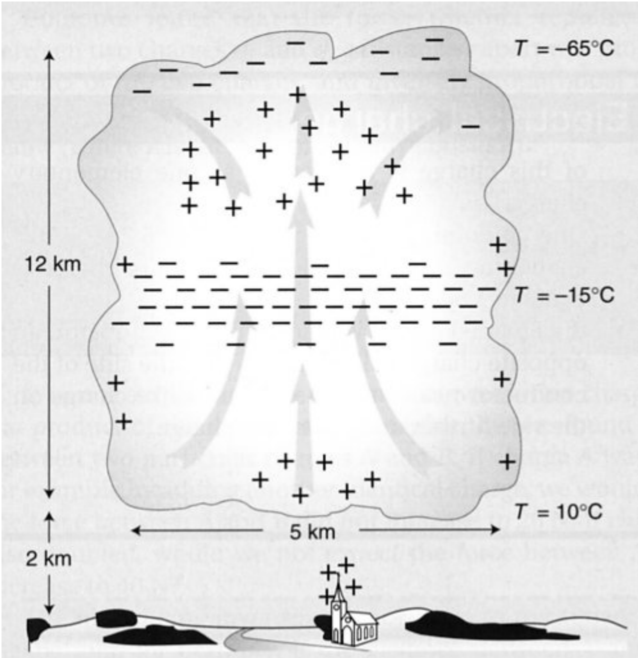


Figure 7.9

A thundercloud can be several kilometres wide and well over 10 km high. Strong updrafts drive the electrical processes that lead to the separation of charge. The strong negative charge will induce positive charges on tall objects on the ground. This may lead to a discharge which can form a conductive path for lightning.

3. **Electrostatic Induction**

When a pen is rubbed on a woollen jumper, it becomes negatively charged. It creates around itself an ***electric field*** in which negative charges will move away from the pen and positive ones towards it.

Any electrons in the paper that are free to move will therefore move to the opposite edge of the paper, leaving the edge closest to the pen with an excess of protons and thus a positive charge.

While the paper is still neutral overall, this positive charge is closer to the pen than is the negative charge on the other edge, and will therefore be attracted more strongly than the negative charge is repelled.

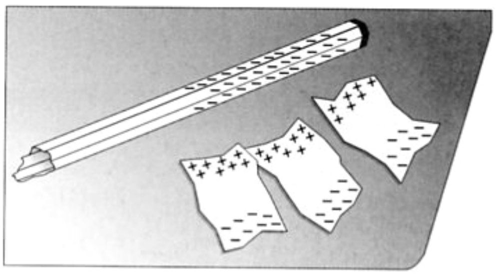


Figure 7.5

A negatively charged pen induces a positive charge on the nearer side of the paper and a negative charge on the opposite side. Because the positive side is closer the paper is attracted.

Fundamental Charge

The elementary charge (e) is given as that charge on a proton or electron.

i.e. **$e = 1.60 \times 10^{-19}$ C** (where C represents the units Coulombs)

Example

a It has been stated that the charge on a rubbed pen would involve many billions of electrons. What is the charge in coulomb carried by 10 billion electrons?

b The charge on a school Van de Graaff generator might be around $-3.0\ \mu C$ ($1\ \mu C = 10^{-6}$ C). How many extra electrons are on the dome?

Solution

a In coulomb, the charge on 10 billion electrons is $10 \times 10^9 \times 1.602 \times 10^{-19} = 1.6 \times 10^{-9}$ C.
This can be referred to as $0.0016\ \mu C$ or as 1.6 nC (nanocoulombs).

b The number of electrons in a charge is the magnitude of the charge divided by the charge on one electron, i.e. $n_e = q/e$.
The negative $3.0\ \mu C$ charge on the Van de Graaff dome consists of

$$n_e = q/e = \frac{3.0 \times 10^{-6}}{1.6 \times 10^{-19}} = 1.9 \times 10^{13} \text{ electrons.}$$

Any normal electrostatic charge involves huge numbers of electrons!

In 1923 Millikan was awarded the Nobel prize for his work on the elementary charge, along with his later experimental work on Einstein's interpretation of the photoelectric effect.

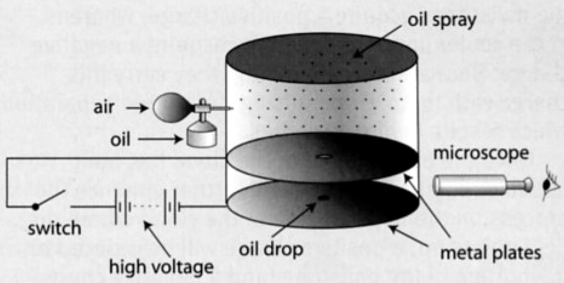


Figure 7.8
Millikan's apparatus consisted of an atomiser which sprayed a fine mist of oil into the chamber. Some of the drops fell into the space between the plates which could be charged via the switch from the battery.

Robert Millikan measured the charge on an electron by finding the electric force on tiny oil drops in a known electric field. If the force on an oil drop was due to the charge of one single extra electron on the drop, and was found to be 8.0×10^{-14} N upwards, what was the strength and direction of the electric field he was using?

Solution

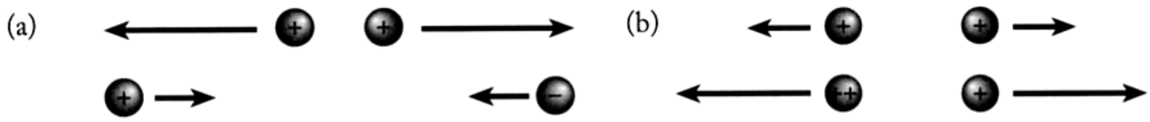
One elementary charge, the charge on an electron, is equal to 1.6×10^{-19} C. The strength of the field is therefore given by:

$$E = \frac{F}{q} = \frac{8.0 \times 10^{-14}}{1.6 \times 10^{-19}} = 5.0 \times 10^5 \text{ N C}^{-1}$$

The direction of the field is downwards. Remember that the direction of the field is the direction of the force on a positive charge. A negative charge experiences a force in the opposite direction to the field. It is interesting to note that Millikan achieved this field by connecting a battery providing 8000 V across two parallel plates 1.6 cm apart.

FORCE BETWEEN CHARGES

Charged objects exert electrostatic forces on each other and act as a distance. Their nature and magnitude depends on the individual charges and their distance apart.



Like charges repel, unlike charges attract. Force is reduced as distance apart is increased

An increase in magnitude of either charge will result in a greater force

Coulomb's Law is used to describe the force between charged objects.

i.e. $$\boxed{F = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{d^2}}$$ **** No calculations are required.**

where F = the force of attraction or repulsion (N)
q_1, q_2 = the charges on the objects (C)
d = the distance between the two objects (m)
e_o = the dielectric constant (permittivity) for the medium ($C^2N^{-1}m^{-2}$)

Example

Two Van de Graaff machines are placed 50 cm apart and switched on. If they both attain a charge of 3 μC, of the same sign, what will be the force between them? (Ignore the size of the machines for the moment.) How would this force change if:

a one of the machines sparks and loses half its charge?
b the machines are moved to a distance of 1 m apart?
c the charges were of opposite sign instead of the same?

Solution

The force between them is given by $F = kq_1q_2/r^2$ where $k = 9 \times 10^9$ N m² C⁻², $q_1 = q_2 = 3 \times 10^{-6}$ C and $r = 0.5$ m.

Thus the force $F = \frac{9 \times 10^9 \times (3 \times 10^{-6})^2}{0.5^2} = 0.3$ N, not a large force, but possibly noticeable. As both machines have charge of the same sign the force will be a repulsive one.

a If one machine loses half its charge we do not need to repeat the calculation, we know that the force will halve to 0.15 N.

b If the distance in an inverse square law doubles, the force will reduce to one quarter. Given the original charge of 3 μC the force will decrease to 0.3/4 = 0.08 N.

c If the charges were opposite, the force would be attractive rather than repulsive. (It is worth noting that if two Van de Graaff machines with the same sign were actually placed this distance apart, the force would probably be less than that calculated because the charges would repel and move to the opposite sides of the domes. On the other hand, if the charges were opposite, they would attract and move to the closer sides of the dome, thus increasing the force.)

ELECTRIC FIELDS

A charged object has an electric field associated with it, and this influences other charged objects placed in it.

The electric field strength is defined as the force per unit charge.

i.e. $$\boxed{E = \frac{F}{q}}$$ **** No calculations are required.**

where
E = the electric field intensity (NC^{-1})
F = the force exerted on the charged object in the electric field (N)
q = the size of the charge on the object in the electric field (C)

Electric field has both size (strength) and direction, hence it is a ***vector***.

The direction of an electric field is taken as the direction a positive test charge, when placed in the field, would move.

i.e. Away from a positively charged object.

The electric field is represented by lines of force or flux.

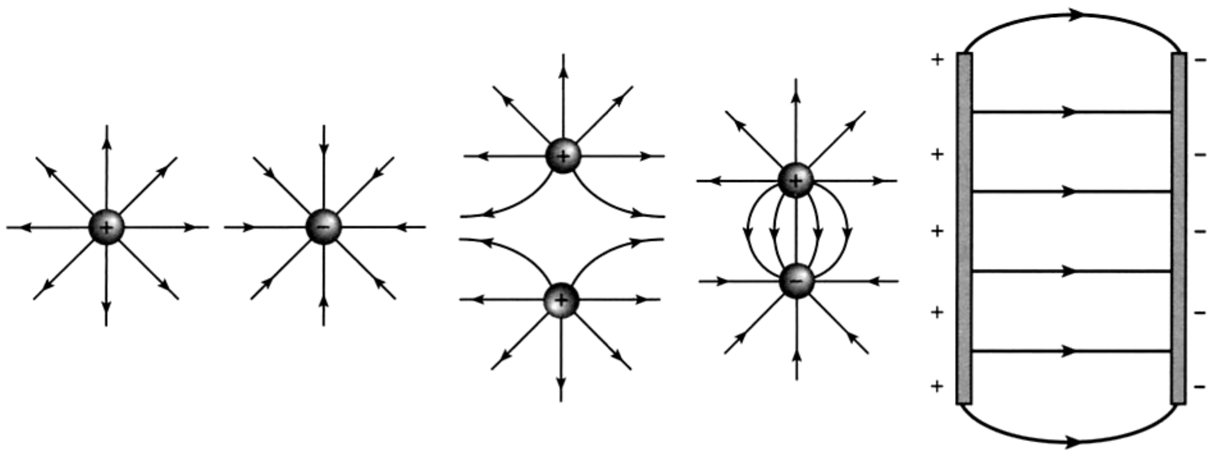


Figure 4.3 Typical electric field patterns.Arrows are always away from positive. Field is strongest where lines of force are closer together.Field is uniform when lines of force are parallel.

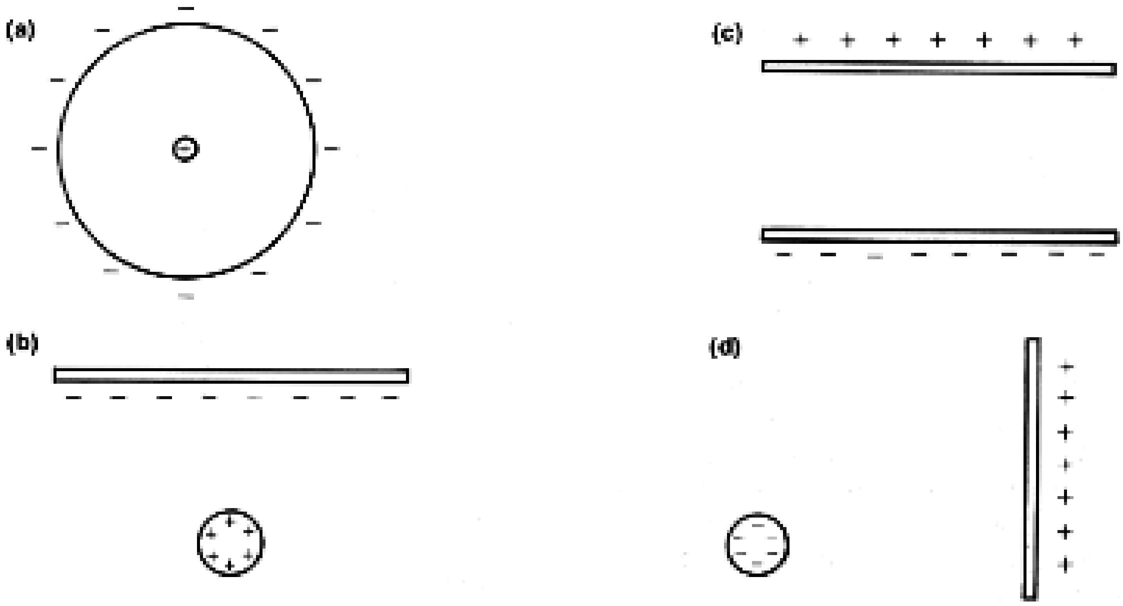
Note

When drawing these diagrams, follow the following "rules".

- Draw the diagrams neatly - no field lines ever touch.
- Field lines enter and leave a surface at right angles.
- Make sure the direction of the field is shown.

Examples

Draw the electric fields associated with the following.



ELECTRIC POTENTIAL

The Earth is defined as having zero potential.

Charged bodies gain electrical potential energy when they are moved in an electric field.

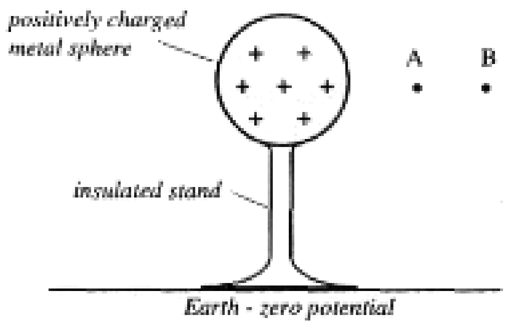
Consider a positively charged body that cannot move and is insulated from the Earth.

If a positive charge is brought to point B, work has to be done to overcome the repulsion.

This work is stored as electric potential energy.

If it is moved to point A, it has a higher electrical potential energy.

Points A and B have different potentials - there is a ***potential difference***.



ELECTRICAL POTENTIAL DIFFERENCE

Moving a charge in an electric field involves work (energy) and hence a change in electrical potential energy.

A potential difference of 1.00 V exists between two points if a charge of 1.00 C moving between those points has its electrical potential energy changed by 1.00 J.

Potential difference is therefore defined as the work done per unit charge in moving the charge through an electric field.

i.e. $$\boxed{V = \frac{W}{q}}$$

where V = the difference in potential between two points in an electric field (V - volts)
W = work done in moving the charge (J)
q = the amount of charge moved in the electric field (C)

Note

The voltage, correctly termed EMF (electromotive force) of an electrical source such as a battery, is a measure of the energy supplied to each coulomb of charge passing through it. Hence a source such as a battery supplies electrical potential energy.

When a charge is placed in an electric field, it experiences a force and moves. It moves from one potential to another, thus requiring some work to be done.

If work is done in moving the charge, it must change E_k.

$$\text{i.e.} \qquad W = \Delta E_k$$

$$\Rightarrow \qquad Vq = \tfrac{1}{2} mv^2 - \tfrac{1}{2}mu^2$$

Since the charge starts from rest, $\tfrac{1}{2} mu^2 = 0$.

$$\therefore \qquad \boxed{\mathbf{W = Vq = \tfrac{1}{2}mv^2}}$$

This Physics principle of accelerating charges is best considered in the electron gun inside a standard TV.

Example

The electrons inside the "red" electron gun of a standard colour TV (using a cathode ray tube) are accelerated by a potential difference of 1.32×10^3 V. Determine how fast these electrons leave the gun to move down the picture tube.

(Data - mass of electron = 9.11×10^{-31}kg, charge on electron = 1.60×10^{-19} C)

$$W = Vq = \frac{1}{2}mv^2$$

$$\Rightarrow \quad (1.32 \times 10^3)(1.60 \times 10^{-19}) = \frac{1}{2}(9.11 \times 10^{-31})v^2$$

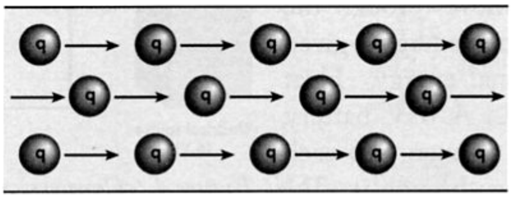
$$\Rightarrow \quad v = 2.153 \times 10^7 \text{ ms}^{-1}$$

$$\therefore \quad \underline{\text{speed} = 2.15 \times 10^7 \text{ ms}^{-1}}$$

ELECTRIC CURRENT

Any material that contains electric charges that are free to move is called a conductor. If such a material is connected between points of different electrical potential, then a flow of charge will occur.

The rate of flow of electric charges, whether positive (+) or negative (-) is called an ***electric current***. In a conducting solution both positive and negative ions are charge carriers while in a metallic conductor only electrons are charge carriers.



- ***Conventional current*** is defined as being in the direction of positive charge flow. It may be considered as a current opposite in direction to a flow of electrons. When we refer to the direction of current in a circuit we are referring to conventional current.

- ***Direct current (DC)*** refers to a current supply where the flow of charge is always in one direction (eg. current from a battery).

- ***Alternating current (AC)*** refers to a current supply where the flow of charge alternates back and forth (eg. household supply).

Figure 7.33
In a wire carrying a DC current the electrons are slowly moving in one direction. In an AC current the electrons hardly move at all, they just vibrate back and forward 50 times a second.

The magnitude of a current is given by:

$$\boxed{I = \frac{q}{t}}$$

where I = current (Amperes - A)
q = amount of charge (C)
t = time taken for the charge to flow (s).

Example

Determine the charge that has flowed through a torch battery producing a current of 3.00×10^2 mA if it has been on for 20.0 minutes.

$I = 0.300$ A $\qquad\qquad I = \dfrac{q}{t}$

$q = ? \qquad\qquad \Rightarrow \quad 0.300 = \dfrac{q}{1.20 \times 10^3}$

$t = 1.20 \times 10^3$ s $\qquad \Rightarrow \quad \underline{q = 3.60 \times 10^2 \text{ C}}$

Physics in action — What really travels along a wire?

It is worth reflecting for a moment on the nature of an electric current in a metal. If it were actually possible to see all the atoms and electrons in a metal wire, we would see a constant blur of activity. The atoms would be vibrating madly and the free electrons would be rushing around at random with enormous speeds—rather like a game of air hockey gone wild! Now imagine that a current suddenly starts to flow in this wire. What difference would it make? The answer is none at all! At least not to any perceptible extent. However, if we could watch a single electron for a few seconds we would notice that, on top of its wild random dance, it had moved a few millimetres in one direction. If you think you might have noticed that, consider that in those few seconds, in its random dance, that same electron would have collided with millions of billions of atoms and covered thousands of kilometres!

If the electrons travel so slowly around the wires, why does the light come on almost instantaneously? In fact, electricity travels along a wire at close to the speed of light. Less than a millisecond after the switch is closed at the power station near Collie, 200 km from Perth, the electrical potential appears at the power line in Perth.

What travels down the power line is not really charge, but a wave of electric field. This wave of electric field pushes the charges a little closer (or a little further apart). It is this change in the concentration of the charges that gives rise to the potential, the voltage, which travels down the wire so quickly.

A simple analogy can be drawn with a garden hose. Provided it is full of water to begin with, water will come out the other end virtually as soon as the tap is turned on. The actual water that goes into the hose when the tap is turned on may take quite a few seconds to emerge from the other end. Again, what travels so quickly down the hose is a pressure pulse, not the actual water. The water then flows because the pressure at the tap end of the hose is higher than that at the other end. Likewise, if the potential at one end of a wire is greater than that at the other, a steady current will flow along the wire.

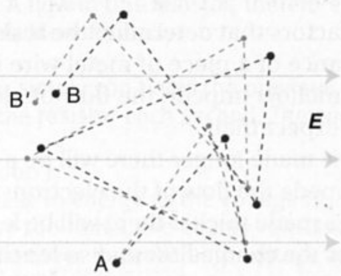


Figure 7.30
This diagram represents the path of an individual electron in a wire as it might appear without (AB) and with (AB′) an electric field ($\boldsymbol{E}$). The motion due to the field, however, has been greatly exaggerated. Drawn to scale it would be almost impossible to notice any difference.

Physics in action — Resistance and the electric field in a wire

We can gain some insight into the nature of resistance from our simple model of the mechanism of charge movement in wires. For a current to flow in a wire, an electric field must be established in order to produce a force on the electrons in the wire. This electric field is set up by the source of EMF which produces a different 'charge concentration' (potential) at one end of the wire relative to that at the other end. This is shown diagrammatically in Figure 7.31.

The concentration of charge along the length will gradually change from being strongly positive at one end to strongly negative at the other end. It is this *changing* concentration which establishes the uniform electric field. At any point an electron 'sees' more positive charge to the left than to the right and will therefore experience a force to the left. (Remember that an electron experiences a force in the direction opposite to the field.) If in fact the field was stronger at one place than another, electrons would move more quickly there and this would weaken the field until it became uniform.

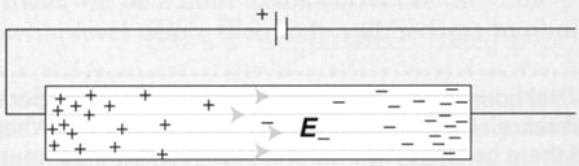


Figure 7.31

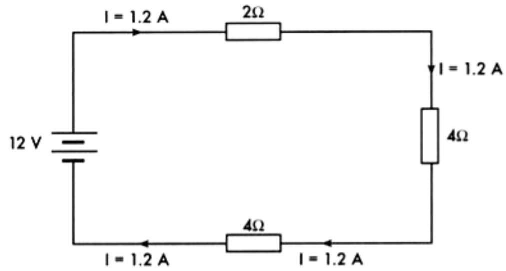
A potential difference applied to the ends of the metal conductor creates an electric field along the length of the conductor.

ELECTRICAL CIRCUITS

Electrical circuits allow the flow of charge from an electrical energy source to components connected within the circuit by conducting material. These circuits maybe very simple or complex.

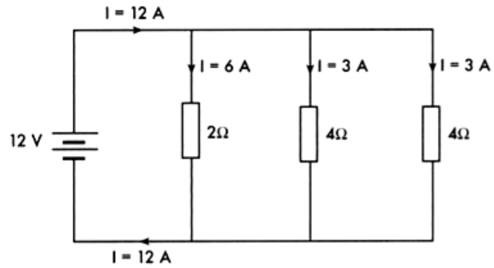
There are three recognisable types of circuits.

- ***Series Circuits***



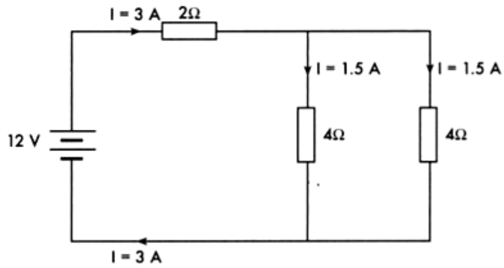
There is only one possible path for the current and the same current flows through all parts of the circuit.

- ***Parallel Circuits***



The current has more than one possible path. The sum of the currents in the different parts of a parallel circuit is equal to the total current .

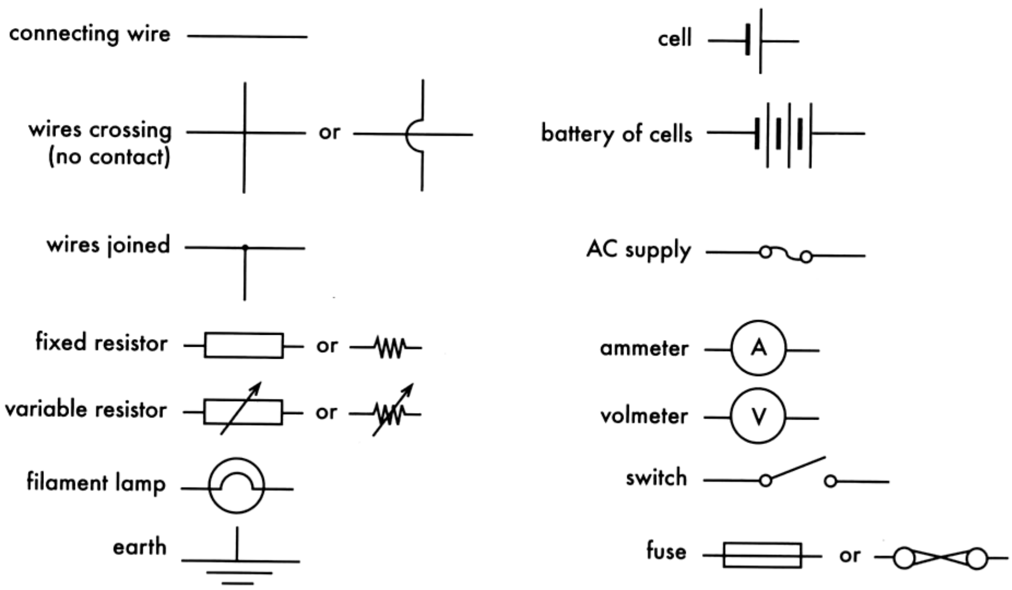
- ***Complex Circuits (Networks)***



These are a combination of series and parallel connections. Most electrical devices involve complex circuits.

CIRCUIT DIAGRAMS

Electrical circuits are best represented by diagrams which show each component of the circuit as a symbol. Conventional symbols used to represent components are shown below.



RESISTANCE

The electrical resistance of materials varies widely, from zero for superconductors to extremely high values for insulators. The availability and number of `free' electrons within a material greatly affects the ease with which a current will flow through it.

The resistivities of different substances (see next page for definition) can be used to classify them as follows.

- ***Superconductors***

Some metals and ceramics have zero resistance when they are cooled to temperatures close to
absolute zero. Applications include magnetic levitation.

Physics file

Superconductors are materials which, at very low temperatures, have zero resistance. Mercury and lead become superconducting at temperatures of 4.2 K and 7.2 K respectively. Curiously, most metals that are good conductors at normal temperatures, gold and copper for example, do not become superconducting at all.

In 1986 a new class of so called 'warm superconductors' was discovered. It was found that various obscure compounds of metal oxides with rare earths became superconducting around the relatively high (but still rather cold!) temperature of liquid nitrogen (77 K). The significance of this was that liquid nitrogen is about as cheap as milk and so it made practical applications look much more feasible.

So far, the applications are mainly in the production of very strong magnets. This is because once an electric current is set up in a superconductor it simply keeps going as there is no resistance to absorb its energy. As you will probably know, a strong electric current creates a strong magnetic field.

- ***Good conductors***

Most metals have low resistivities (in the order of 10^{-8} Ωm) and are suitable for use in electrical wiring. Metals and alloys with relatively higher resistivities (about 10^{-6} Ωm) are useful for resistor and heater filaments.

- ***Semi-conductors***

Non-metals such as germanium and silicon have relatively high resistance. However the resistance of these materials can be markedly and selectively reduced by introducing specific impurities.

This special property allows the production of transistors and microchips, which are vital components for the electronics industry.

- ***Insulators***

Materials such as wood, glass and plastic have extremely high resistivities (10^{10} to 10^{15} Ωm) and are therefore very useful insulators. They are very necessary for the safe operation of electrical devices and the prevention of electrocution.

POTENTIAL DIFFERENCE AND CURRENT - OHM'S LAW

The relationship between the current flowing through a resistor and the potential difference applied to it was first investigated by the German physicist Ohm. He found that:

> "provided that the temperature remains constant, the current through a resistor is proportional to the potential difference applied across it".

This is known as Ohm's Law. This law only applies to metals and carbon at constant temperature. It is usually expressed as follows.

$$\boxed{V = IR}$$

where V = voltage or potential difference (V)
I = current (A)
R = resistance (Ohms - Ω)

Example

A resistor of 5 Ω is supplied with a potential which can vary from 1 V to 100 V.

a What will be the range of current that will flow in it?
b How much energy will be dissipated in the resistor each second?

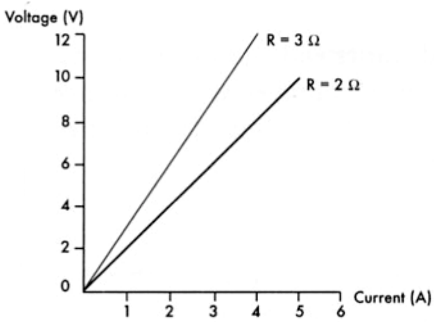
Solution

a 5 V are required to make 1 A flow in this resistor, therefore at 1 V the current will be $^1/_5$ A or 0.2 A. More formally:
At 1 V, $I = V/R = {}^1/_5 = 0.2$ A
At 100 V, $I = {}^{100}/_5 = 20$ A (or simply 100 times the previous answer).

b At 1 V, 0.2 C flow through the resistor each second. The energy is given by $E = qV = 0.2 \times 1 = 0.2$ J.
At 100 V, $E = 20 \times 100 = 2000$ J
Notice the very large increase in energy as the voltage is increased. As the voltage increased by 100 times, the energy released each second increased by 10 000 times because both voltage and current increased by 100 times.

Ohmic Conductors

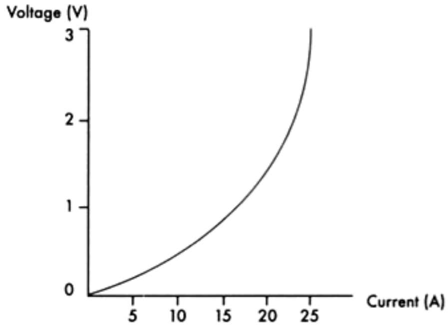
Ohmic conductors are those whose resistance remains constant as the current is varied. The slope of their *V/I* graph would be constant. Resistors and conducting wires used in circuits are ohmic conductors within specific limits.



(b) Ohmic conductors.
Gradient $\frac{V}{1}$ is constant.

Non-Ohmic Conductors

Non-Ohmic conductors do not show a constant relationship between voltage and current. Their resistance can increase with temperature, as does a globe filament, or decrease, as is the case with thermistors.



(c) Non Ohmic conductors. Typical torch globe characteristics.

VOLTMETERS AND AMMETERS

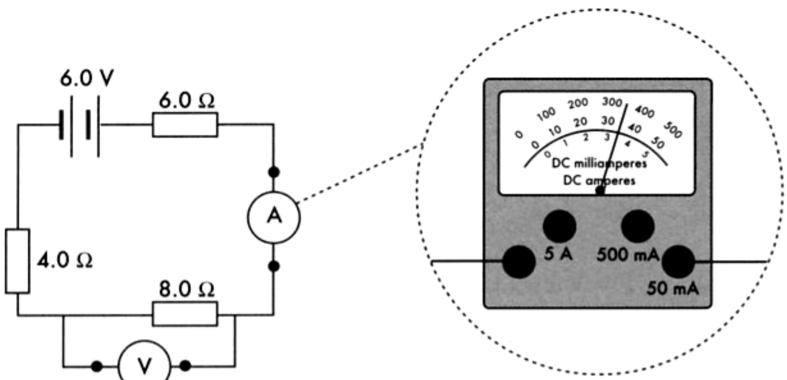
A ***voltmete***r is used to measure the voltage (potential difference) between two points. It is therefore connected ***in parallel*** to the component it is measuring the voltage across.

Voltmeters are high resistance instruments. This is so that very little current is diverted from the circuit when a voltage measurement is made.

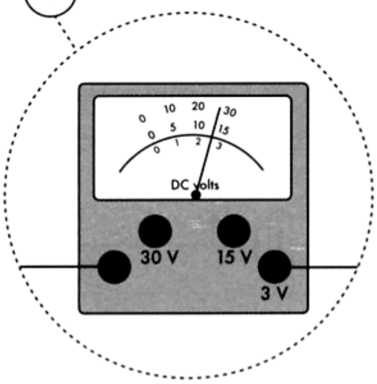
An ***ammeter*** is used to measure the current flowing through a particular point in a circuit. It is therefore connected ***in series*** at that point in the circuit.

Ammeters are very low resistance instruments so that they will have little effect on the current flowing in a circuit.

Polarities of connections. The positive terminal of an ammeter or voltmeter is always connected to the part of the circuit coming from the positive terminal of the power supply.



An **ammeter** is always connected in series in a circuit. An ammeter has one black terminal and usually two or three red terminals. The black terminal is connected to the part of the circuit diagram closest to the negative power supply. One of the red terminals is selected depending on scale needed. Always choose 5A scale first to avoid damage to meter.



A **voltmeter** is always connected in parallel in a circuit. Similarly to the ammeter it has one black terminal (connected to the part of the circuit closest to the negative of the power supply) and two or three red terminals for different scales

RESISTANCE AND RESISTIVITY

The electrical resistance of a conductor depends upon the type of material it is made from, as well as its size and shape. Factors that affect the resistance are:

- ***Length (l)*** - resistance increases proportionately with the length of a conductor.

  i.e. $R \propto l$

- ***Cross-sectional area (A)*** - the greater the cross-sectional area of a conductor, the smaller its resistance.

  i.e. $R \propto \frac{1}{A}$

- ***Nature of the conducting material*** - an electrical property of materials is their ***resistivity (ρ)***. The resistance of a conductor is proportional to its resistivity.

  i.e. $R \propto \rho$

- ***Temperature*** - the higher the temperature of a metal the greater its resistance. The higher the temperature of carbon the lower its resistance.

These first three relationships can be combined as follows.

$$\boxed{R = \frac{\rho l}{A}}$$

**** No calculations are required.**

where R = resistance (Ω)
r = resistivity (Ωm)
l = length (m)
A = cross-sectional area (m^2)

Example

Normal household wiring uses 1.8 mm diameter copper wire. What is the resistance of a 10 m long piece of this copper wire? What voltage drop will there be along it if a current of 10 A is flowing through it?

Solution

The resistivity of copper is found from Table 7.3: $\rho = 1.7 \times 10^{-8}\ \Omega$ m.
The cross-sectional area of the wire is given by:
$A = \pi r^2 = 3.14 \times (0.9 \times 10^{-3})^2 = 2.5 \times 10^{-6}$ m²
The resistance is therefore found from:
$R = \rho L/A = 1.7 \times 10^{-8} \times 10/(2.5 \times 10^{-6}) = 0.068\ \Omega$
If the wire is carrying a 10 A current there will be a voltage drop of
$V = IR = 10 \times 0.068 = 0.68$ V along the length of the wire.

Table 7.3 Resistivity of some common materials at 20°C

Material	Resistivity, ρ (Ω m)
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminium	2.8×10^{-8}
Tungsten	5.5×10^{-8}
Tungsten (2000°C)	70×10^{-8}
Nichrome	100×10^{-8}
Doped silicon*	1×10^{-3}
Pure silicon	3×10^{3}
Pure water	5×10^{3}
Soils and rock	$10^{3} \rightarrow 10^{7}$
Wood	$10^{8} \rightarrow 10^{11}$
Glass	$10^{10} \rightarrow 10^{14}$
Fused quartz	10^{16}

ELECTRICAL ENERGY

Sources of electricity such as batteries provide energy to the charges flowing in a circuit. A battery contains stored chemical energy that is given to the electrons as electrical potential energy. This is converted to kinetic energy as the electrons are attracted to the positive terminal.

Since the electrons collide with atoms as they move through the different parts of the circuit, their energy is given up to the atoms as vibrational energy. Heat and sometimes light results.

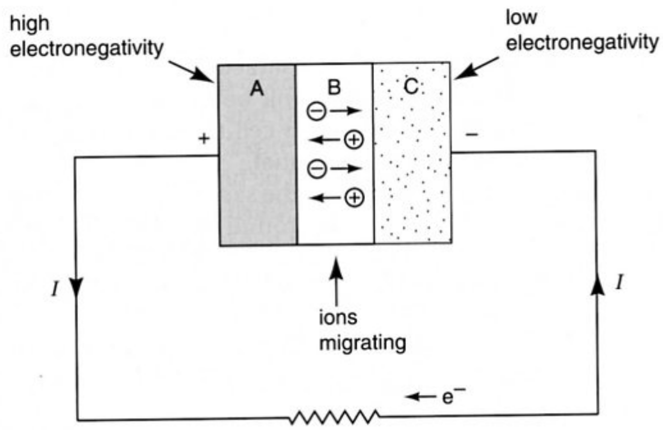


Figure 7.51
A cell consists of two materials that attract electrons to different degrees, along with a 'go-between' which acts as a conductor. In this diagram material A attracts electrons most (high electronegativity) and material C least. Material B acts as the go-between.

e.g. A 6.00 V battery will supply 6.00 J of energy to each coulomb of charge that passes through it (from $W = Vq$). This exact energy is then lost to the external circuit by each coulomb of charge.

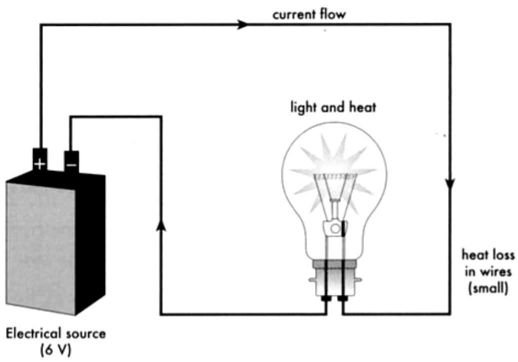


Figure 4.6 Conversion of electrical energy to heat and light. Note: conventional current flow is taken as from positive to negative.

ELECTRICAL POWER

Power is defined as the rate of doing work or releasing energy.

$$P = \frac{W}{t} = \frac{Vq}{t} = VI \quad (\text{since } I = \frac{q}{t})$$

i.e. $$\boxed{P = VI}$$

Examples

1.

Two different torch bulbs are rated as 2.8 V, 0.27 A, and 4.2 V, 0.18 A.

a Which will be the brightest?

b Could they be interchanged?

Solution

a The brightness is indicated by the power used—although only about 4% becomes light energy. $P = VI$ and so their powers are $2.8 \times 0.27 = 0.76$ W (or 760 mW) and $4.2 \times 0.18 = 0.76$ W. The bulbs will be the same brightness.

b Although the power of each bulb is similar, using them in the wrong torch will either result in the bulb burning out or running dim. The bulbs are designed to work at a certain voltage. If a greater voltage is used, too much current will flow in the bulb which will result in it burning out.

2.

A motor car's two headlights are each rated at 50.0 W and operate on a 12.0 V power supply. Calculate

(a) The current flowing in each headlight when they are in use.

(b) The charge passing through each globe every second.

(c) The total energy consumed by the two headlights during a 2.00 hour night journey.

$P = 50.0$ W each light
$V = 12.0$ V
$I = ?$
$q = ?$
$t = 2.00$ h

(a) $P = VI$

$$I = \frac{P}{V} = \frac{50.0}{12.0} = 4.17 \text{ A (for each light)}$$

(b) $q = It$
$= (4.17)(1)$
$= 4.17$ C

(c) $E = Pt$
$= (50)(2)(2.0 \times 60 \times 60)$
$= 7.20 \times 10^5$ J

From V = IR and P =VI, the power dissipated in a component in a circuit is given by:

$$\boxed{P = VI = I^2R = \frac{V^2}{R}}$$

Electrical Energy Used

The energy consumed by an electrical appliance depends upon the rate of energy use (power rating), and the time for which it is operating.

$$E = Pt = VIt = I^2Rt = \frac{V^2t}{R}$$

The Kwh - A Handy Unit

Household electrical energy is measured in units called "kilowatt-hours" (kWh) since it is a more convenient unit. Your electricity bill from Western Power will refer to electricity units used. These are kWh units.

Since 1 kW = 1000 Js and 1 h = 3600 s

$\therefore$ **1 kWh = 3.60 x 10^6 J = 3.60 MJ**

Examples

1.

How much energy does a 100 W light bulb use in half an hour?

Solution

Here $P = 100$ W and $t = 0.5$ h.
So $E = 100$ W × 0.5 h = 50 W h.
This could also be given as 100 W × 1800 s = 180 000 J = 180 kJ.

2.

What is the current drawn by a 1500 W electric kettle if it operates on a 240 V supply? How much electrical energy (Joules) will it use if it is on for 4.50 minutes? Calculate the cost of the energy used if Synergy charge 12.67¢ per kWh.

P = 1500 W   I = ?
V = 240 V   E = ?
t = 4.5 min   Cost = ?
= 270 s

(a) P = VI

$\therefore I = \frac{P}{V} = \frac{1500}{240} = 6.25$ A

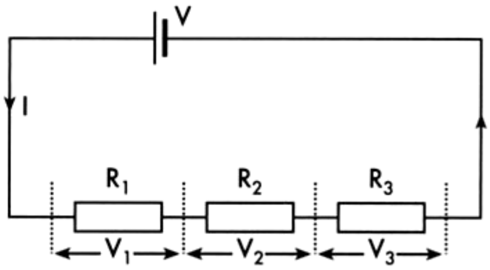
(b) E = Pt
= (1500)(270)
= 4.05×10^5 J

(c) 1 kWh = 3.6×10^6 J
$\therefore$ No. of units of electricty used (kWh)
$= \frac{4.05 \times 10^5}{3.6 \times 10^6} = 0.113$ kWh

$\therefore$ Cost = (units) × (12.67¢)
= (0.113) × (12.67¢) = 1.43¢

SERIES CIRCUITS

The diagram at right shows resistors that are connected in series.



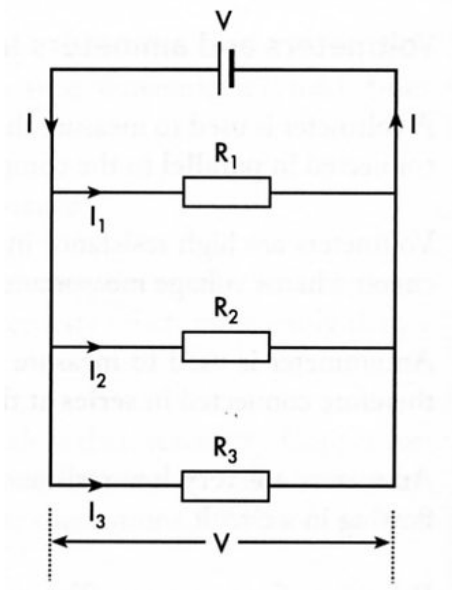
- The supply voltage (V) is shared among the resistors.

  ie. $V_T = V_1 + V_2 + V_3$

- The same current (I) goes through all the resistors.
- The total resistance is the sum of the individual resistances.

  i.e. $\boxed{\mathbf{R_T = R_1 + R_2 + R_3}}$

Parallel Circuits



The diagram at right shows resistors that are connected in parallel.

- The current is shared among the resistors.

  i.e. $I_T = I_1 + I_2 + I_3$

- The same voltage (V) exists across each resistance.
- The total resistance is less than any of the resistances and is calculated as shown below.

$$\boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Examples

1.

When a globe was connected to a 12 V battery it was found that the current flowing was 2.0 A. Calculate the globe's (a) resistance, (b) power rating.

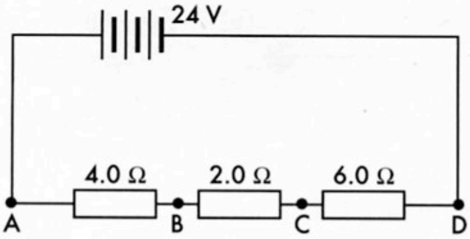
V	=	12 V	(a)	$R = \frac{V}{I} = \frac{12}{2.0} = 6.0\ \Omega$
I	=	2.0 A		
R	=	?	(b)	$P = V\,I = (12)(2.0) = 24\ W$
P	=	?		

2.

Three resistors, 4.0 Ω, 2.0 Ω, and 6.0 Ω are connected in series to a 24 V supply as shown. Determine: (a) total resistance, (b) the current flowing in each resistor, (c) the voltage across each resistor.

(a) $R_T = R_1 + R_2 + R_3$
$= 4.0 + 2.0 + 6.0$
$= 12\ \Omega$

(b) $V = 24\ \text{V}$ $\quad I_T = \dfrac{V}{R_T} = \dfrac{24}{12}$
$I = ?$
$R_T = 12\ \Omega$ $\quad = 2.0\ \text{A}$



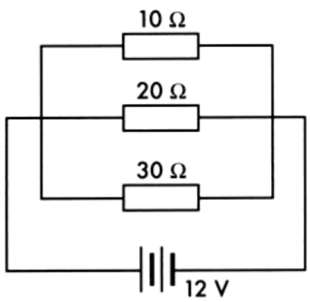
(c) $V_{AB} = I\,R = (2.0)(4.0) = 8.0\ \text{V}$
$V_{BC} = I\,R = (2.0)(2.0) = 4.0\ \text{V}$
$V_{CD} = I\,R = (2.0)(6.0) = 12.0\ \text{V}$

Note that the voltages across the three resistors add up to the supply voltage, 24 V. The same current goes through each resistor.

3.

Three resistors, 10 Ω, 20 Ω, and 30 Ω are connected in parallel to a 12 V supply as shown. Determine: (a) total resistance, (b) the current in each resistor.

(a)
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$
$$= \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$
$$= \frac{6+3+2}{60} = \frac{11}{60}$$
$$\therefore R_T = \frac{60}{11} = 5.45\ \Omega$$



(b) Each resistor has a potential difference of 12 V across it.

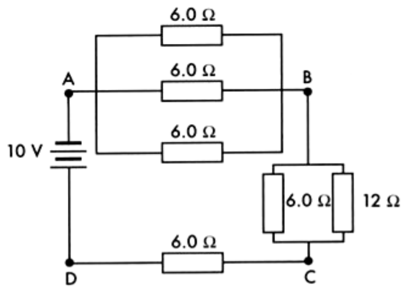
$\therefore$ for 10 Ω resistor $\quad 1_{10\,\Omega} = \dfrac{V}{R} = \dfrac{12}{10} = 1.2\ \text{A}$

similarly $\quad 1_{20\,\Omega} = \dfrac{V}{R} = \dfrac{12}{20} = 0.60\ \text{A}, \qquad 1_{30\,\Omega} = \dfrac{V}{R} = \dfrac{12}{30} = 0.40\ \text{A}$

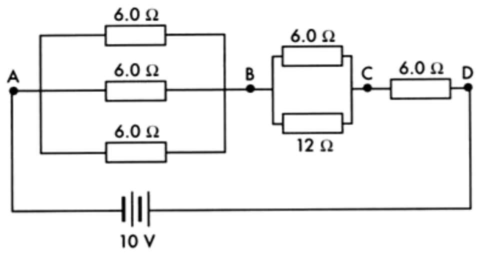
Note that $I_T = 1.2 + 0.60 + 0.40 = 2.2$ A. Can you verify this?

Complex Circuits

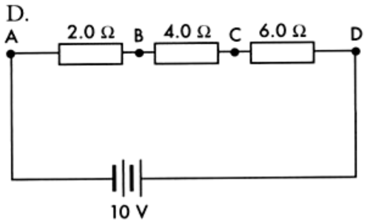
Most electrical circuits are a combination of series and parallel circuits. To determine the equivalent resistance of the circuit, each part of the circuit is dealt with separately and simplified to a series circuit. The diagram below shows how a network of resistors can be simplified to a single equivalent resistor.



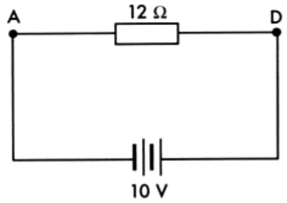
(a) Complex network consisting of 2 parallel networks, A to B and B to C, in series with a 6 Ω resistor, C to D.



(b) The same circuit, redrawn for clarity.

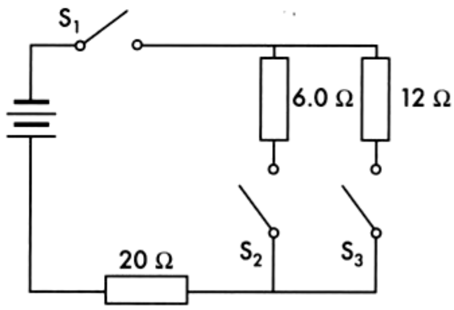


(c) Using $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ ... etc a single resistor replaces the parallel networks



(d) The equivalent resistance for the circuit.

Example



Three resistors are connected to a DC power supply as shown. When all switches are closed it is found that the current in the 20 Ω resistor is 500 mA.

(a) For these conditions determine:
  (i) The total resistance of the circuit.
  (ii) The potential difference across each resistor.
  (iii) The current in the 6 Ω resistor.
  (iv) The voltage of the DC supply.

(b) If S_3 is opened and S_1 and S_2 remain closed determine the current now flowing through the 20 Ω resistor.

(a) (i) The 6.0 Ω and 12 Ω resistors are in parallel

$$\frac{1}{R_p} = \frac{1}{6.0} + \frac{1}{12} \quad \text{*}R_p = \text{resistance of parallel section}$$

$$= \frac{2.0 + 1.0}{12} = \frac{3.0}{12} \therefore R_p = \frac{12}{3.0} = 4.0\ \Omega$$

Hence $R_T = 20 + 4 = 24\ \Omega$

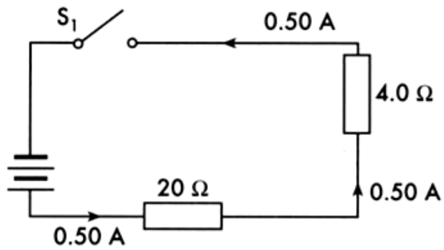
(ii) A simplified circuit shows the parallel section as 4.0 Ω.

Using $V = IR$

$V_{20\Omega} = (0.50)(20) = 10\ V$

$V_{4\Omega} = (0.50)(4) = 2.0\ V$

The voltage across the 6.0 Ω and 12 Ω resistor is 2.0 V. Across the 20 Ω resistor it is 10 V.



(iii) Since the 6 Ω resistor has 2.0 V across it

$$I_{6\,\Omega} = \frac{V}{R} = \frac{2.0}{6.0} = 0.33\ A$$

(iv) The voltage of the supply is equal to the total of the voltages across the resistors in series.

$V = 10 + 2.0 = 12\ V$

Note: Taking the 12 Ω resistor out of the parallel section actually increased the overall resistance of the circuit (Why?). This caused a drop in the current through the circuit.

(b) If S_3 is opened the total resistance for the circuit will be

$R_T = 20 + 6.0 = 26\ \Omega$

∴ Current flowing in this series circuit

$$I = \frac{V}{R} = \frac{12}{26} = 0.46\ A$$

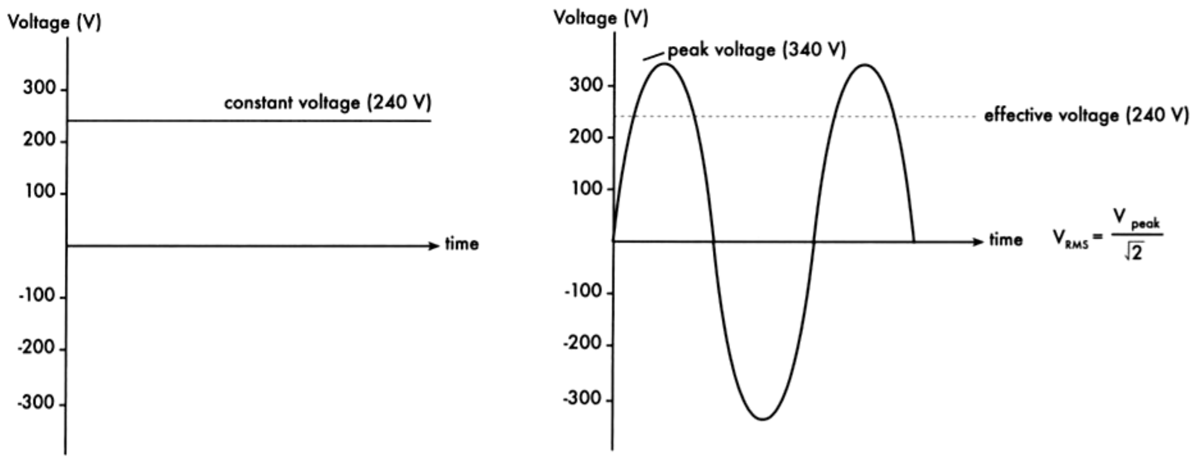
AC POWER SUPPLY

The power generated for both homes and industry is alternating current (AC). This form of power has many distinct advantages over a DC (direct current) supply.

AC power

- More easily generated.
- Generally safer and cheaper for most applications.
- Easily transformed to higher or applications lower voltages is suitable for devices such as clocks and tape recorders since its frequency can be precisely controlled.
- Can be transmitted at high voltages to minimise energy losses.

The nature of AC power means that both the voltage and current continually rise and fall and vary in direction with time. The effective voltage (or current) is a mathematical average called the root mean square voltage (V_{RMS}).



DC voltage (b) AC voltage

ELECTRICAL SAFETY

Effects Of Current On Humans

When an electric shock occurs, it depends on the resistance of the part of the body affected.

e.g. Outer skin has a high resistance (poor conductor) while internal tissues underneath have a much lower resistance (better conductor).

Wet or sweaty skin has a much lower resistance again, resulting in even better conduction.

Muscles are the most sensitive to electric currents, given that the electrical impulses sent by nerves from the brain control their movement. Any outside currents can over-ride these electrical impulses and cause the muscle to contract into spasm.

i.e. The heart can go into ventricular fibrillation (or even stop) during electrocution where it ceases to beat rhythmically.

The effects on the human body are summarised below.

Current (mA)	Effect
1	Threshold of feeling.
10 - 16	Paralysis of some muscles - can't let go.
16 - 80	Plus pain, possible fainting.
> 80	Plus heart may go into ventricular fibrillation.
> 1000	Heart may restart if current removed quickly; severe burns.

When the heart has stopped or is in ventricular fibrillation, it is externally started again by using "paddles" to send a large current through it. This causes the heart to "kick start" and/or resume its normal rhythm.

Safety Devices

1. ***Fuses***

These are designed to melt when a current that is above the desired maximum (e.g. 10 A) flows in a household circuit.

This could occur if a power board (or single socket with double adaptors "piggy-backed") is overloaded with too many devices. The more devices placed in parallel, the lower the resistance and higher the current drawn from the board. If this total current exceeds 10 A, then the wiring could heat up and melt, causing a fire. The fuse wire melts first and protects any wiring.

2. ***Circuit Breakers***

These detect if the current has exceeded the maximum and switch off to protect the wiring. They can easily be reset. They are equivalent to fuses in their performance.

3. ***Earth Wire***

Three pin plugs have an earth wire that is connected externally from the house to a water pipe by a wire. All devices within the house have a wire leading to this connection with the earth.

Should the casing of a device become “live” due to a short-circuit, the earth wire acts as a short-circuit to the ground, causing a large current to flow to earth and tripping the appropriate fuse or circuit breaker.

4. ***Double Insulation***

Appliances with double insulation are made largely of plastic and the active and neutral wires are double-wrapped in plastic. The plugs on these devices have only two pins - there is no need for an earth wire.

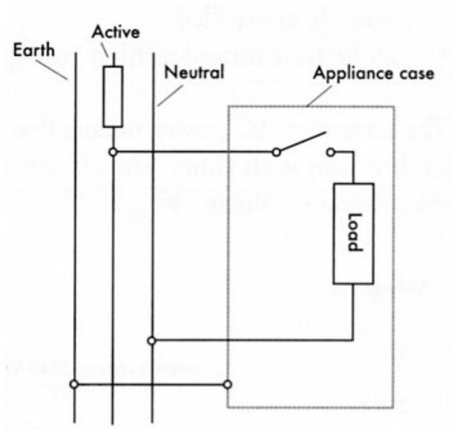
Double insulation is ***not*** a foolproof method of preventing shocks. Nevertheless it is better protection than an earth wire.

5. ***Earth Leakage Circuit Breakers (ELCB)***

Also known as ***residual current circuit breakers (RCCB)***, these devices monitor the current in both the active and neutral wires (which should be the same). If the current in one drops slightly due to a “leakage” of current (as in a short circuit), it disconnects the circuit in less than 30 ms. This is 6-8 times faster than a fuse and is far safer, resulting in the person not even detecting the current in many cases and hence avoiding a shock.

It has been a law in WA for many years for new homes to have ELCB’s in their meter boxes.

6. ***Earth Wire***



The earth wire is connected to the metallic case of an appliance so that should the appliance become live, the current will flow harmlessly to earth . A fuse will most likely blow.

7. ***Double Pole Switches***

These connect (and disconnect) both the active and neutral wires when the switches / switch is operated . This is preferable to the action of the single pole switches (DPST) which could, through incorrect wiring, leave an appliance effectively connected to the mains even though the switch is off.

Plugs

There are two types.

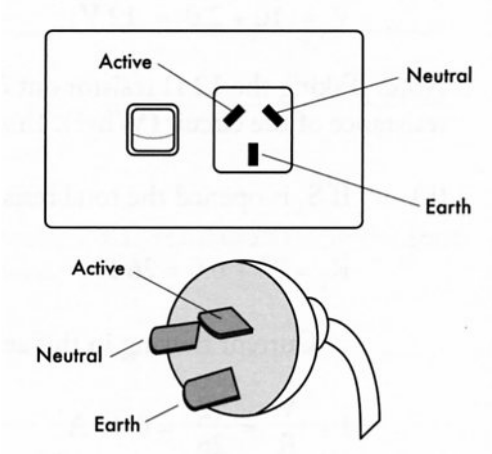
1. ***Three Pin***

   This has three wires connected to it.

   Active - carries the current to the appliance (***brown*** in colour).

   Neutral - return wire to the switchboard or meter box (***blue***).

   Earth Wire - provides a connection to earth in case of short circuits (***green and yellow***).



2. ***Two Pin***

   This has only the active and neutral wires. No earth wire is required as these are only used for double-insulated appliances.